

Ryota Nakamura Faculty of Data Science, Musashino University Tokyo 135-8181, Japan ryota.nakamura@ds.musashino-u.ac.jp

June 8, 2026

To the Editor-in-Chief and Editorial Board, *IEEE Access*

Re: Submission of “Cache the Known, Ask Only the Novel: A Confidence-Routed Harness for Sub-1B Language Models”

Dear Editors,

Please consider the attached manuscript for publication in *IEEE Access*. The work studies how to deploy sub-1-billion-parameter (“tiny”) language models reliably and cheaply on device, and reaches a conclusion that we believe is both practical and counter to common practice.

Problem and contribution. The usual route to making a tiny model useful is to strengthen the model (fine-tuning, larger checkpoints). We argue and demonstrate the opposite for tiny-model *systems*: the path to better performance, cost, and reliability is not a stronger model but a better *division of labour* — separate inputs into “known” and “novel,” answer the known with deterministic, non-parametric components, and spend the language model only on the genuinely novel. Concretely, we pair a frozen sub-1B model with an experience cache of verified cases that exposes two complementary paths (a no-LLM embedding classifier and the LLM conditioned on retrieved exemplars and distilled rules), and we add a *confidence gate* that decides between them.

Why it is a contribution, not a heuristic. The central engineering result is that the routing decision is principled and validated: the non-parametric classifier’s own confidence predicts whether its no-LLM answer is correct at mean AUROC 0.83 across seven datasets (nearest-neighbor similarity alone is near chance). Routing by that signal matches or beats the best single path (85.6% vs 83.9% no-LLM, 57.0% LLM) while invoking the language model on only ~4% of inputs, and approaches an oracle ceiling. A comprehensive sweep — 11 datasets $\times$ 8 sub-1B models (88 model–dataset cells) — shows the no-LLM path dominates every LLM-path variant (71.9% vs 26–29% mean), is nearly invariant to class count (80% from 2 to 77 classes), and rivals LoRA fine-tuning at zero training cost; structuring the model’s own failures into discriminative rules helps the LLM path only marginally. We also delimit the claim honestly: a generative-QA pilot marks where the no-LLM path stops transferring, so all claims are scoped to closed-set classification, with open-ended generation and reasoning left to future work.

Suitability and significance for IEEE Access. The paper offers a clear, reproducible engineering message for the large practitioner audience deploying small models at the edge — *when not to call the model* — supported by a large empirical study, validity controls (leave-one-theme-out splits, label-corruption and 1-NN baselines, Wilson confidence intervals, paired McNemar tests), an external re-test on six real-world datasets, and a full open-source release of code, scripts, and result data.

Originality and ethics. This manuscript is original, has not been published previously, and is not under consideration elsewhere. There are no conflicts of interest. All datasets used are public benchmarks or author-created; no human-subjects or private data are involved. The author has approved the submission.

Thank you for your consideration. I look forward to the reviewers’ feedback.

Sincerely,

Ryota Nakamura Faculty of Data Science, Musashino University